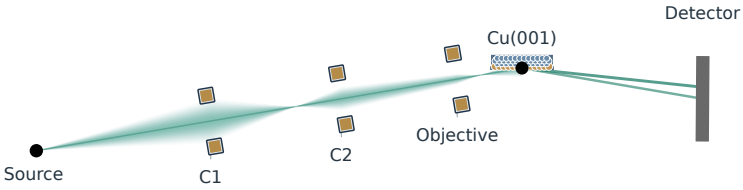
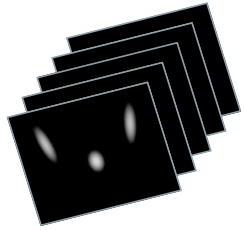
a Initial microscope guess

Unknown source, lens positions and focal lengths



b Record diffraction patterns

Vary lens powers and translate the detector



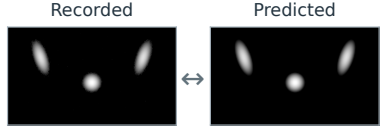
Known relative settings · unknown absolute geometry

Measured data

Predict patterns

c Fit the microscope

Compare measurements with model predictions



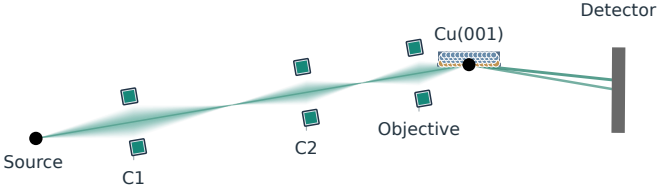
Update positions and lens strengths

Joint fit across the recorded image series

Fit

d Recovered microscope

Fitted geometry with uncertainty estimates



Reflection calibration

Illustrated with simulated Cu(001) diffraction data; beam envelopes enlarged